EDS 223: Homework Assignment 3

West Coast Marine Aquaculture

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1. Load in libraries

```
library(tidyverse)
library(sf)
library(terra) # raster handling
library(tmap) # map making
```

2. Load in and prepare data

```
# Load sea surface temperature data
sst_2008 <- rast("data/average_annual_sst_2008.tif")
sst_2009 <- rast("data/average_annual_sst_2009.tif")
sst_2010 <- rast("data/average_annual_sst_2010.tif")
sst_2011 <- rast("data/average_annual_sst_2011.tif")
sst_2012 <- rast("data/average_annual_sst_2012.tif")

# Load ocean depth data
depth <- rast("data/depth.tif") # raster

# Load EEZ data
eez <- st_read("data/wc_regions_clean.shp") # vector

# Combining SST rasters into a raster stack
sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012) # raster
```

3. Process SST and depth data so that they can be combined

```
# Average SST from 2008-2012 and convert from C to Kelvin
avg_sst_C <- mean(sst_stack, na.rm = TRUE) - 273.15

# Resample depth to match res and extent of SST
depth_resampled <- resample(depth, avg_sst_C, method = 'bilinear')

# Checking and converting depth to match SST resolution, extent, and CRS
if (all(res(avg_sst_C) == res(depth_resampled))) {
  print("The resolutions match!")
} else {
  warning("Update resolutions to match!")
  res(depth_resampled) <- res(avg_sst_C)
}
```

```
[1] "The resolutions match!"
```

```
if (ext(avg_sst_C) == ext(depth_resampled)) {
  print("The extents match!")
} else {
  warning("Update extents to match!")
  ext(depth_resampled) <- ext(avg_sst_C)
}
```

```
[1] "The extents match!"
```

```
if (crs(depth_resampled) == crs(avg_sst_C)) {
  print("The coordinate reference systems match!")
} else {
  warning("Update coordinate reference systems to match!")
  crs(depth_resampled) <- crs(avg_sst_C)
  # if (crs(depth_resampled) == crs(avg_sst_C)) {
  #   print("The coordinate reference systems now match!")}
}
```

4. Function to find suitable areas for different species

```r
suitable_areas <- function(
  min_sst,
  max_sst,
  min_depth,
  max_depth,
  species_name
) {

  ## Find suitable locations
  # Reclassifying SST and depth data into suitable locations
  # SST reclassification matrix
  rcl_sst <- matrix(
    c(
      -Inf, min_sst, 0,
      min_sst, max_sst, 1,
      max_sst, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
  )

  # Reclassifying sst
  sst_zones <- classify(avg_sst_C, rcl = rcl_sst)

  # Depth reclassification matrix
  rcl_depth <- matrix(
    c(
      -Inf, max_depth, 0, # should max_depth be negative?
      max_depth, min_depth, 1,
      min_depth, Inf, 0
    ),
    ncol = 3,
    byrow = TRUE
  )

  # Reclassifying depth
  depth_zones <- classify(depth_resampled, rcl = rcl_depth)

  # Finding locations that satisfy both SST and depth conditions
  # Adding oyster zone rasters
  sst_depth_zones <- c(sst_zones, depth_zones)
```

```r
# Finding locations that satisfy both SST and depth conditions
suitable_locs <- lapp(sst_depth_zones, function(sst, depth) sst * depth)

## Determine the total suitable area within each EEZ
# Making EEZ crs equal oyster_suitability CRS
sf::st_crs(eez) <- terra::crs(suitable_locs)

# Converting sf object to terra SpatVector
eez_vect <- terra::vect(eez)

# Rasterizing EEZ data
eez_raster <- rasterize(eez_vect, suitable_locs, field = 'rgn_id')

# Calculating area of each cell
cell_sizes <- cellSize(suitable_locs, unit = "km", mask = TRUE)

# Making sure only cells classified as 1 contribute to area totals
suitable_area <- mask(cell_sizes, suitable_locs, maskvalues = 0)

# Selecting suitable raster cells that fall within each EEZs region
suitable_area_by_eez <- zonal(
  suitable_area,
  eez_raster,
  fun = 'sum',
  na.rm = TRUE
)

## Mapping EEZ regions colored by amount of suitable area
# Converting zonal results to a data frame
suitable_area_df <- as_tibble(suitable_area_by_eez)

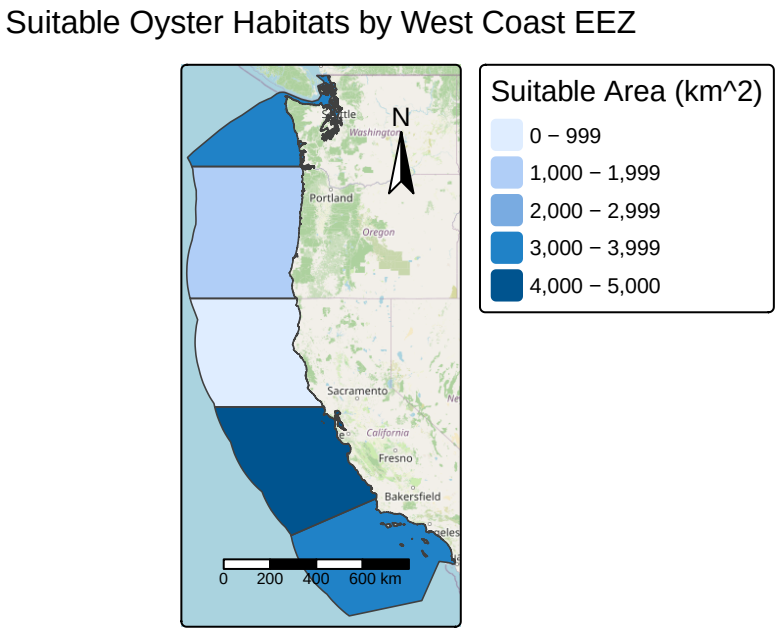
# Joining suitable area to EEZ polygons
eez_map <- left_join(eez, suitable_area_df, by = "rgn_id")

# Creating map
tm_shape(eez_map) +
  tm_basemap("OpenStreetMap") +
  tm_fill(
    fill = "area",
    fill.legend = tm_legend(title = "Suitable Area (km^2)")
  ) +
  tmap_options(component.autoscale = FALSE) +
```

```
    tm_borders(lwd = .8) +
    tm_title(paste(
      "Suitable",
      species_name,
      "Habitats by West Coast EEZ"),
      size = 1
      ) +
    tm_scalebar(text.size = .5, position = c("bottom", "left")) +
    tm_compass(position = c("top", "right"))
}
```

Testing the function for oysters

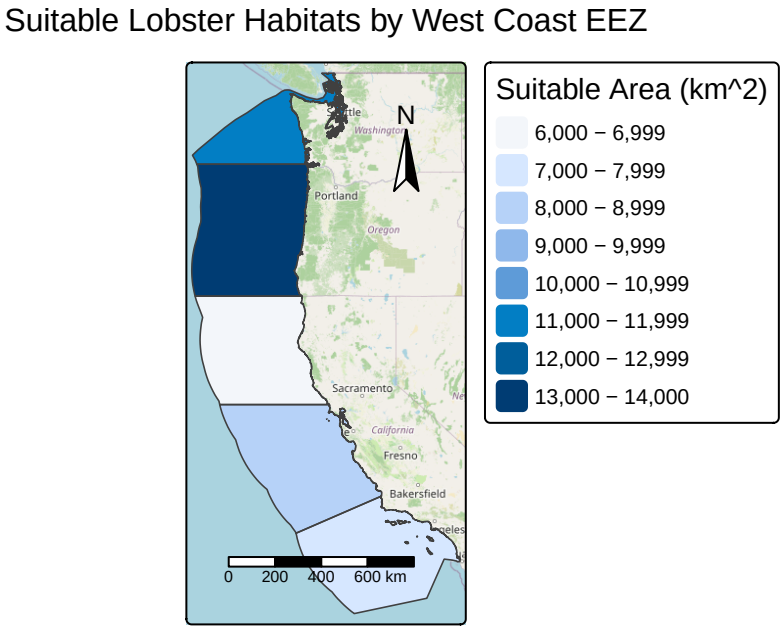
```
suitable_areas(11, 30, 0, -70, 'Oyster')
```



The optimal growth conditions for oysters are 11-30 deg C and 0-70 m below sea level. The resulting map shows that the Central California EEZ has the greatest amount of suitable area for the growth of these oysters, with a total of 4,374 km^2. Northern California has the least amount of suitable area, with a total of 324 km^2.

Testing the function for California spiny lobsters

```
suitable_areas(10, 21, 0, -150, 'Lobster')
```



The optimal growth conditions for California spiny lobster are 10-21 deg C and 0-150 m below sea level. The resulting map shows that the Oregon EEZ has the greatest amount of suitable area for the growth of these lobsters, with a total of 13,578 km^2. Northern California has the least amount of suitable area, with a total of 6,870 km^2.